11/24/55

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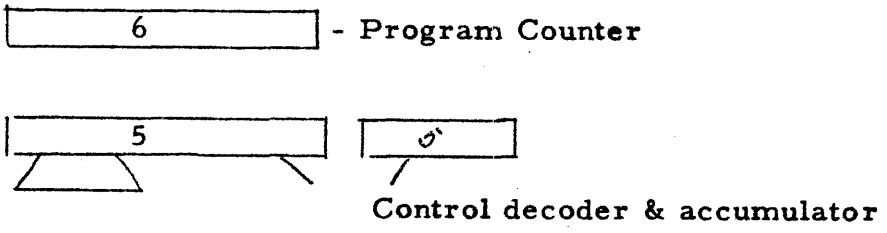
Project Stretch

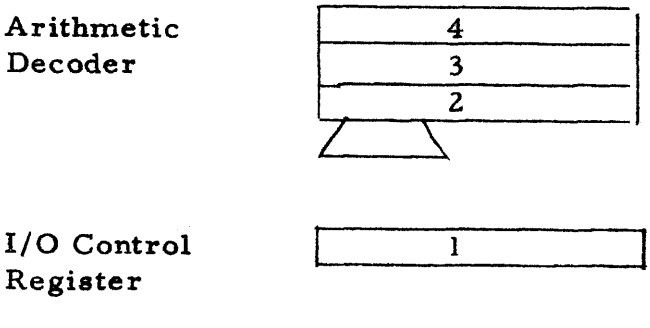
Logical rules for sequencing memory references:

1. An order of priority is assigned to registers requiring memory references. This priority is:

   a) Input/output control register
   b) Arithmetic decoding registers - highest priority given to arithmetic instruction nearest in time to being executed.
   c) Control decoding register including the control accumulator.
   d) Program counter.

e.g.





2. The order of priority is effectively altered by supression of references from the various registers by the rules given below:

   a) Data references called for by the instructions in the arithmetic decoder are suppressed if the memory box referred to is in use, or if the arithmetic instruction being executed is "long" relative to a memory cycle. Example - when a multiply instruction is being executed, references from arithmetic decoder suppressed and the con-

trol decoder takes precedence after the I/O control register.

b) When the control decoder is occupied and instruction not executable the program counter will discontinue calling for memory references.

3. As an arithmetic instruction goes through the control decoder signals are generated to control memory references and are sent with the instruction to the arithmetic decoder. Instructions are classed as "long" or "not long", requiring a memory reference to "read" or to "write", and referring to a memory box with a "long" or a "short" cycle. If an instruction in the arithmetic decoder is a "long" operation, references by instruction following the "long" instruction are inoperative.

   Only a "read" from "slow" memory may be executed before the instruction reaches box 2 of the arithmetic decoder.

4. Memory references are inhibited or interchanged in the control decoder. If the instruction "(X)" in the control decoder refers to a memory location that is also referred to by an instruction in the arithmetic decoder, the instruction "X" will be retained in the control decoder until the earlier instruction is executed. An exception, to this inhibition, is effected when the instruction to load from memory location Q is immediately preceded by an instruction to store in location Q. Under these conditions the two cycles "write" and "read" are replaced by one cycle "write" and retain data. When an arithmetic "read" operation is immediately preceded by an arithmetic "write" and the instructions refer to different locations, the memory references are interchanged.

5. When a transfer instruction is brought into the control decoder it is executed immediately, provided the decision is not dependent on the status of the main arithmetic unit. If the outcome of the instruction is dependent on the status of the main arithmetic unit, the instruction is held in the control decoder until a decision can be made.

6. The main arithmetic unit generates 4 types of signals to control timing while executing an instruction:
   a) Permissible to start "read" from "slow" memory [1]
   b) Permissible to start "read" from "fast" memory.
   c) Permissible to start "write" to "fast" memory.
   d) End of operation.

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1. "Write" to "slow" memory signal will be part of b, or c.